AMAZON BASE BUILDING COMPLIANCE STANDARD

Task Title	Hot water services
APM Object	CT.WATER.HOTSERVICE
APM Description	Hot water storage and supply systems
General tasks always required	Follow safety procedures - e.g. 1MRA, LOTO, PPE, etc. Ensure that all WOs are carried out by qualified, certified staff with appropriate familiarisation and supervision as required Add notes for each observation and create a FWO if necessary to eliminate defects Use appropriate tools, chemicals and industry standard practices (RME policy permitting) Ensure relevant logs, readings and certification is uploaded to APM prior to WO closure. CT evidence must be additionally uploaded to the DMS (WorkDocs in DE) To suggest improvements please raise a SIM or contact document authors Leonardo Vitale (LEOVITAL) Rob Gisby (EWARTGXR) or APM, Benito Sanchez Sanchez de Puerta (BPPUERTA) SIM - Link to SIM template

Last Update: (Version)	ago-23	**Last Update Change Detail:**	Transfer to new template. Rationalisation of EU RME Minimum standards. Alignment with dependant PM. Released as Draft.	**Activities may be covered by No-Go restrictions:**	Not Specified	**Special Certification Required:**	**Delivered By:**
Evidenced part or full by PM:	Not Specified	**Related BBM PM:**	HVAC.HTR	**Related CT Areas:**	CT.WATER.LEGIONELLA - CT.WATER.COLDSERVICE - CT.SAFE.WASH		

Scope

Any site with hot water services, such as hot water supply storage tanks, calorifiers, mixer taps, showers and distribution network, that might lead to a risk of employees contracting legionnaires disease.
Legionella risk assessment requirements are covered under CT.WATER.LEGIONELLA. Cold water systems are covered in CT.WATER.COLDSERVICE.
Additional requirements of emergency showers and A are covered under CT.SAFE.WASH

Using this document

This document is intended as a guide and may not be exhaustive. Whilst this document is based extensively on professional guidance and subject to periodic review, individual sites must be diligent to ensure they are aware of, and comply to, local and changing regulations.
The format of these standards is intended to improve navigation between maintenance and compliance standards and to allow frequent, minor updates to a single living resource. Please avoid offline or printed versions and check version dates and change log.
In the event of becoming aware of errors or omissions in this document, sites must provide detail, including references to relevant source information to their local planner for escalation - SIM Link above.
This document is considered released as draft.
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Maintenance and Compliance Relationship

Where relevant, associated Base Building PM's are listed on the header . In many cases maintenance contributes to, or fulfils compliance requirement. This may not be exhaustive and is intended to assist users to identify relevant maintenance.
If a BBM Maintenance activity fully meets the compliance requirements below, periodic upload of PM evidence will be required to APM and DMS / Workdays depending on region.

Key to Abbreviations

The key to the below document is as follows:
CT - Amazon compliance / statutory task - not otherwise covered by a BBM Standard PM. Evidence to be provided into APM and DMS upon task closure.
CT* - This is a conditional task based or based on findings from previous inspections. If a frequency is not used, a further appropriate frequency needs to be in either the region, or 'ALL'. Local teams must ensure appropriate frequency.
(CT) - Compliance task which is likely met or exceeded by a CT task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met.
PM - Requirement can demonstrated through appropriate maintenance - Refer to relevant PM Standard
(PM) - Maintenance task which is met or exceeded by a PM task in [ALL] and therefore achieved by proxy BUT local site must satisfy themselves that requirements are fully met including regional / manufacturer requirements.
EVID - Periodic upload of evidence to DMS, for non-CT task lines. For example, evidence of PM completion, certification, licence etc.
EVID* - Conditional requirement to upload evidence at the interval provided
N A R - No Additional Requirement

Required Tasks

Region	Task Index	Task Frequency	Task Requirements	Qualification of Inspector	Records Required	PM	Weekly	Monthly	3 Monthly	6 Monthly	12 Monthly	24 Monthly	Additional
All	1	Based on risk assessment but at least monthly	Check temperatures at in flow and return at calorifiers and storage cylinders to ensure 50-60°C These locations should be identified in the Legionella Risk Assessment.	Competent person, usually sourced internally.	Legionella log book (results of temperature checks, management actions etc) Legionella Risk assessment Schematic diagram of all hot water services			CT					
All	2	Based on risk assessment but at least monthly	Check water temperature up to one minute to see if it has reached 50°C in the sentinel taps. These locations should be identified in the Legionella Risk Assessment.	Competent person, usually sourced internally.	Legionella log book (results of temperature checks, management actions etc) Legionella Risk assessment Schematic diagram of all hot water services			CT					
All	3	Based on risk assessment but at least monthly	Check outlets associated with each combination water heater to ensure the unit stores water at 50-60°C. These locations should be identified in the Legionella Risk Assessment.	Competent person, usually sourced internally.	Legionella log book (results of temperature checks, management actions etc) Legionella Risk assessment Schematic diagram of all hot water services			CT					
All	4	Based on risk assessment, but at least every 3 months.	Check temperatures in subordinate recirculating loops ensuring at least 50°C is achieved. These locations should be identified in the Legionella Risk Assessment.	Competent person, usually sourced internally.	Legionella log book (results of temperature checks, management actions etc) Legionella Risk assessment Schematic diagram of all hot water services				CT				
All	5	Based on risk assessment, but at least every 3 months.	Dismantle, clean and descale shower heads and hoses (including emergency showers)	Competent person, usually sourced internally.	Legionella log book (results of temperature checks, management actions etc) Legionella Risk assessment Schematic diagram of all hot water services				CT				

All	6	Based on risk assessment, but at least every 6 months.	Check the temperature of the water provided by point of use heaters (<15l) ensuring at least 50°C is achieved	Competent person, usually sourced internally.	Legionella log book (results of temperature checks, management actions etc) Legionella Risk assessment Schematic diagram of all hot water services					CT			
All	7	Based on risk assessment, but at least every 6 months.	Flush through and purge to drain all expansion vessels.	Competent person, usually sourced internally.	Legionella log book (results of temperature checks, management actions etc) Legionella Risk assessment Schematic diagram of all hot water services					CT			
All	8	Based on risk assessment, but at least annually	Arrange for samples to be taken from hot water calorifiers and storage cylinders, in order to note condition of drain water and identify internal scale or sludge	Competent person, best practice to outsource . Any analysis should be completed at UKAS accredited laboratory.	Inspection records, report or certificate. Legionella Risk assessment Laboratory testing results Schematic diagram of all hot water services						CT		
All	9	Based on risk assessment, but at least annually	Visual check on internal surfaces of calorifiers and storage cylinders for scale and sludge. Check representative taps for temperature as per [2] on a rotational basis.	Competent person	Inspection records, report or certificate. Legionella Risk assessment Schematic diagram of all hot water services						CT		
All	10	Based on risk assessment, but at least annually	Inspect the header units of each combination water heater.	Competent person	Inspection records, report or certificate. Legionella Risk assessment Schematic diagram of all hot water services	PM					EVID		
All	11	Based on risk assessment, but at least annually	Undertake service of all Thermostatic Mixing Valves (TMV's) to the manufacturer's instructions.	Competent person	Inspection records, report or certificate. Legionella Risk assessment Schematic diagram of all hot water services	PM					EVID		

Related Regional Legislation

Region	Relevant legislation	Vendor Certification	Statutory requirement
ALL	No region specific legislation specified	N/A	N/A
CZ	Act No. 258/2000 Coll. Decree 73/2010 Coll., ČSN 331500, ČSN 333320 ed.2, ČSN 332570, ČSN 332000, ČSN EN 62305, ČSN 331600.ed.2, Decree 50/1978 Coll.	Accredited company Competent technician	Carryout, as a producer of hot water, regular analyses according to the extent covered by the legislation. Frequency determined by the Protocol on the list of external influences, between 6 and 60months.
DE	Trinkwasserverordnung (TrinkwV) Bekanntmachungen und Empfehlungen des Umweltbundesamtes zur TrinkwV DVGW W 551 - Trinkwassererwärmungs- und Trinkwasserleitungsanlagen - Technische Maßnahmen zur Verminderung des Legionellenwachstums - Planung, Errichtung, Betrieb und Sanierung von Trinkwasser-Installationen VDI/DVGW 6023 - Hygiene in Trinkwasser-Installationen - Anforderungen an Planung, Ausführung, Betrieb und Instandhaltung	Accredited specialist company (Zugelassener Fachbetrieb)	Visual inspection of hot water storage facility at least annually to identify any physical defects. Visual inspection of the drinking water heater every two months (check the temperature setting of the device, it should be at least 60 ° C.) Maintenance and cleaning of hot water heater every 12 months. Maintenance of softening unit every 6 months. Maintenance of Pressure reducer every 12 months. Maintenance of system separator and backflow preventer: maintenance and function check every 12 months.
ES	RD 140/2003, which establishes the health criteria of the quality of water for human consumption. RD 865/2003, laying down the hygienic-sanitary criteria for the prevention and control of legionellosis. RD 1027/2007, which approves the Facilities Regulation Thermal in Buildings (RITE) Building Technical Code (CTE). - Health Standards Basic Document: water supply (HS4)	RITE: Competent person and authorised Maintainer. Legionella inspections: authorised companies (by ENAC).	Annual inspection, cleaning, disinfection and sampling of equipment. Legionella periodic inspections and sampling: Cooling towers and evaporative condensers: annual inspections. Domestic hot water systems with storage tank and return circuit: every 4 years. Industrial humidifying plants: every 2 years.
FR	Public Health Code – Art. R1321-43 to 61 https://www.legifrance.gouv.fr/codes/id/LEGIARTI000006909561/2007-01-12	Qualified person	Regular inspection of header units of each water heater in line with risk assessment but at least annually.

IT	Legislative Decree 04/09/2008 n.81, Annex IV Legislative Decree 02/02/2001 n.31, Annex I Guidelines for prevention and control of legionnaires' disease Min salute 2015 UNI EN 918:2014(hot and cold-water supply and distribution systems). UNI EN 806-5:2012 (operation and maintenance of installations inside buildings for the conveyance of water intended for human consumption) UNI EN ISO 11731: 2017 (cultural methods for the determination of the charge in water parameters)	Qualified person	Control measurements are required. Frequency periodical of check is not specified by Italian legislation; however the best practices, in accordance with the guidelines of the Ministry of health regarding the water system maintenance are: - Descaling of less worn elements by immersion in acid solution (sulfamic acid, acetic acid, etc.) and subsequent disinfection (not less than 30 minutes, in cold water containing at least 50 mg/L free chlorine). - Periodically replacement of faucet valve, faucet water filters, shower heads and pipes, as well as any other discontinuity element. The frequency of replacement is usually a function of water characteristics. - Microfiltering that allows the removal of Legionella from the water exiting at the point of use, through a mechanical barrier.
PL	Regulation of Health Minister of 7 December 2017 on the quality of water intended for human consumption (Journal of Laws 2017, item 2294) Regulation of Minister of Infrastructure of 12 April 2002 on the technical conditions to be met by buildings and their location (Journal of Laws 2019, item 1065) Act of 20 July 2017 on Water Law (Journal of Laws 2021, item 624)	The State Sanitary Inspection	Regular samples to be taken from calorifiers and storage cylinders not stipulated by regulations Check calorifier and storage cylinder's flow and return water temperature not stipulated by regulations. Check water temperature at sentinel taps (up to 1 minute) in line with risk assessment, but at least annually. Visual checks of internal surfaces of calorifiers and storage cylinders for scale and sludge and check temperature on representative taps in line with risk assessment, but at least annually. Regular inspection of header units of each water heater in line with risk assessment but at least annually. Check temperature of water heater outlets not stipulated by regulations Check temperature of subordinate recirculating loops not stipulated by regulations Check temperature of the water provided by point of use heaters (if <15L) not stipulated by regulations
ROI	Safety, Health and Welfare at Work Act 2005 (irishstatutebook.ie)	Competent person, best practice to outsource to specialist	Regular samples to be taken from calorifiers and storage cylinders in line with risk assessment, but at least annually. Check calorifier and storage cylinder's flow and return water temperature in line with risk assessment but at least monthly. Check water temperature at sentinel taps (up to 1 minute) in line with risk assessment, but at least monthly. Visual checks of internal surfaces of calorifiers and storage cylinders for scale and sludge and check temperature on representative taps in line with risk assessment, but at least annually. Regular inspection of header units of each water heater in line with risk assessment but at least annually. Check temperature of water heater outlets in line with risk assessment but at least monthly.
SK	Act No. 355/2007 Coll. on the protection, promotion and development of public health and on the amendment of certain laws Act No. 124/2006 Coll. on safety and health at work Ordinance No. 247/2017 Coll. on details of drinking water quality, drinking water quality control, monitoring and risk management program for drinking water supply Ordinance No. 508/2009 Coll. on details for ensuring safety and health protection when working with technical pressure, lifting, electric and gas equipment and laying down technical equipment which is considered to be the reserved technical equipment	Competent and certified person	Annual water quality control or control based at request from the regional public health office. Frequency of inspection depends on type of equipment and its parameters (performance and pressure). Inspection of equipment is realized at least twice a year. .
UK	The Control of Substances Hazardous to Health Regulations 2002 Health and Safety at Work etc. Act 1974 (legislation.gov.uk) Health and Safety at Work (Northern Ireland) Order 1978 (legislation.gov.uk) Legionnaires' disease: The control of legionella bacteria in water systems. Approved Code of Practice and guidance on regulations L8 (hse.gov.uk) HSG 274 Part 2 - The control of legionella bacteria in hot and cold water systems (legionellacontrol.com)	Competent person, best practice to outsource to specialist	Regular samples to be taken from calorifiers and storage cylinders in line with risk assessment, but at least annually. Check calorifier and storage cylinder's flow and return water temperature in line with risk assessment but at least monthly. Check water temperature at sentinel taps (up to 1 minute) in line with risk assessment, but at least monthly. Visual checks of internal surfaces of calorifiers and storage cylinders for scale and sludge and check temperature on representative taps in line with risk assessment, but at least annually. Regular inspection of header units of each water heater in line with risk assessment but at least annually. Check temperature of water heater outlets in line with risk assessment but at least monthly.